

Short-Term Rainfall Prediction Applied to an Urban Basin

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1. Introduction

Urban flooding in rapidly responding basins poses a major challenge for hydrometeorological monitoring systems due to the occurrence of short-duration convective precipitation events [1]. This work presents a machine learning approach for short-term rainfall prediction using pluviometric time series collected from automatic rainfall stations distributed across an urban basin [2, 3].

2. Methodology

2.1 Study Area

The basin of the Mburicaó stream, in Asunción, Paraguay, is a dense urban area vulnerable to flash floods due to the increase in impervious surfaces.



Figure 1: Automatic rainfall stations and Mburicaó basin boundary, Asunción, Paraguay.

2.2 Rainfall Event Detection

Rainfall events were defined using a minimum threshold of 0.3 mm/10 min and a maximum dry interval of three consecutive samples.

2.3 Feature Extraction

Accumulation features were extracted from rainfall measurements recorded 10–30 minutes earlier at local and neighboring stations.

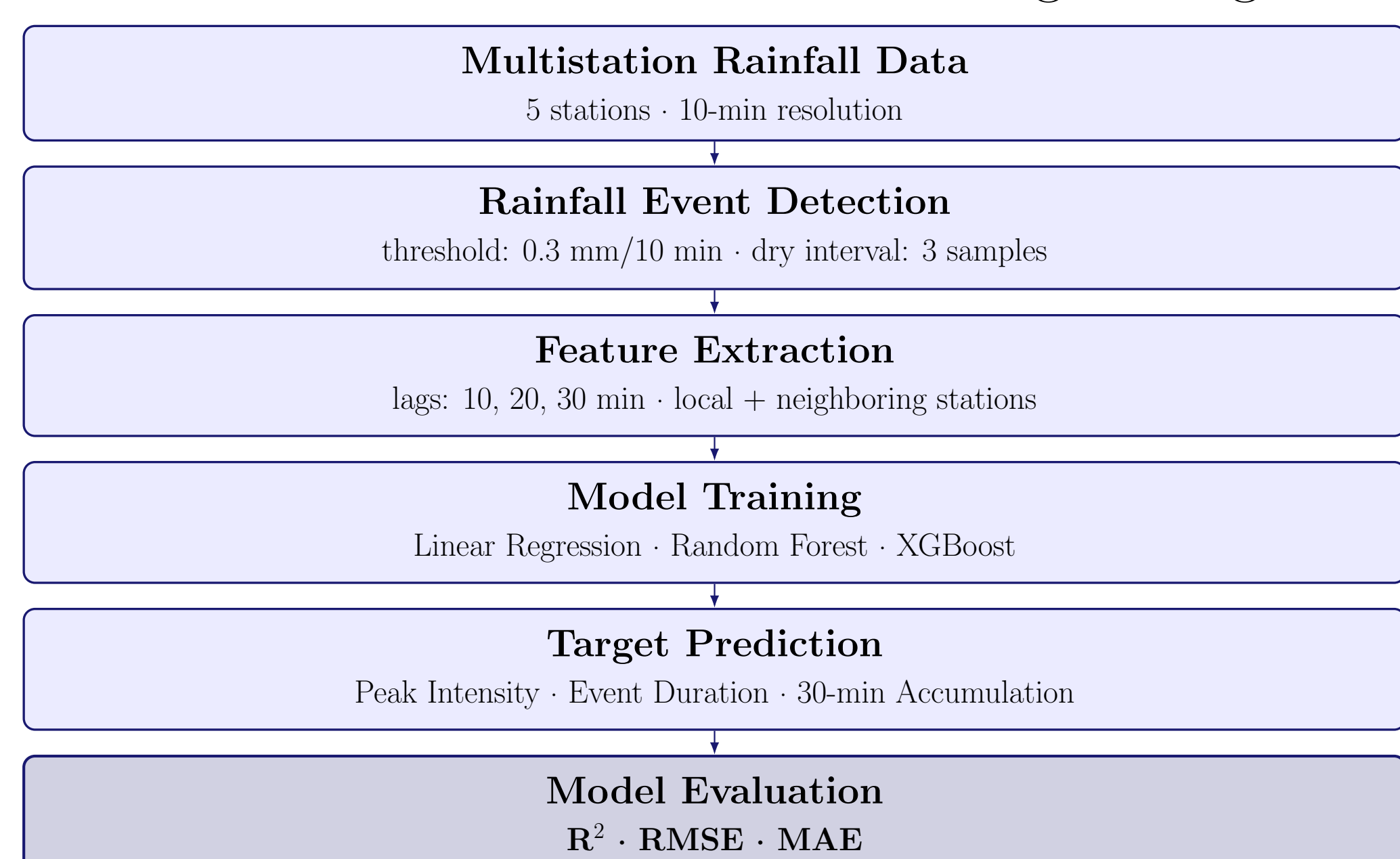


Figure 2: Methodology pipeline for short-term rainfall prediction.

3. Results

Tree-based models outperformed linear regression for all three target variables, indicating nonlinear rainfall dynamics.

Model	Target Variable	R^2
XGBoost (multistation)	30-min accumulation	≈ 0.56
Random Forest	Peak intensity	≈ 0.73
Linear Regression	30-min accumulation	< 0.40

Table 1: Performance comparison.

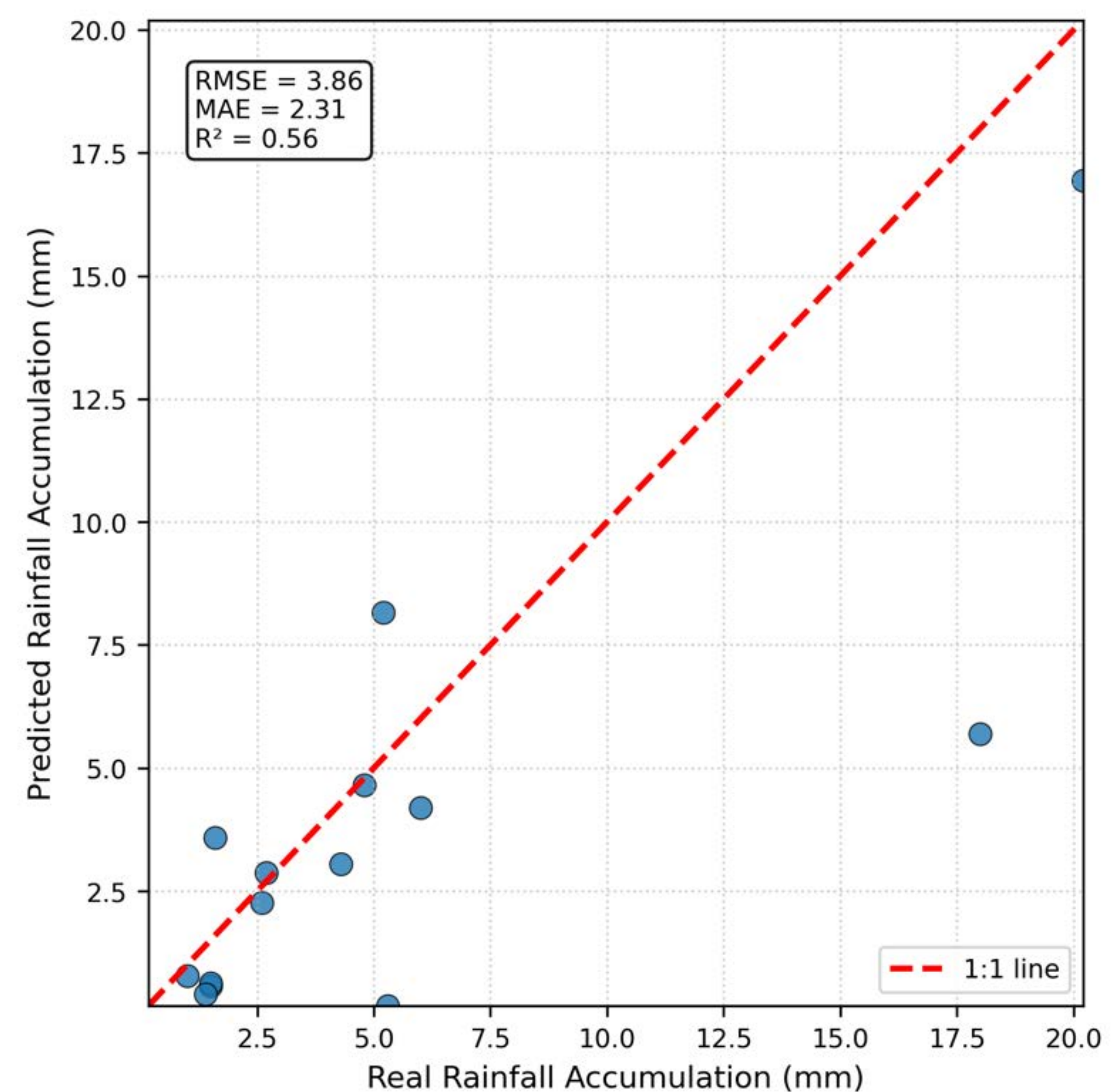


Figure 3: XGBoost prediction of 30-minute accumulation (RMSE = 3.86, MAE = 2.31, $R^2 = 0.56$).

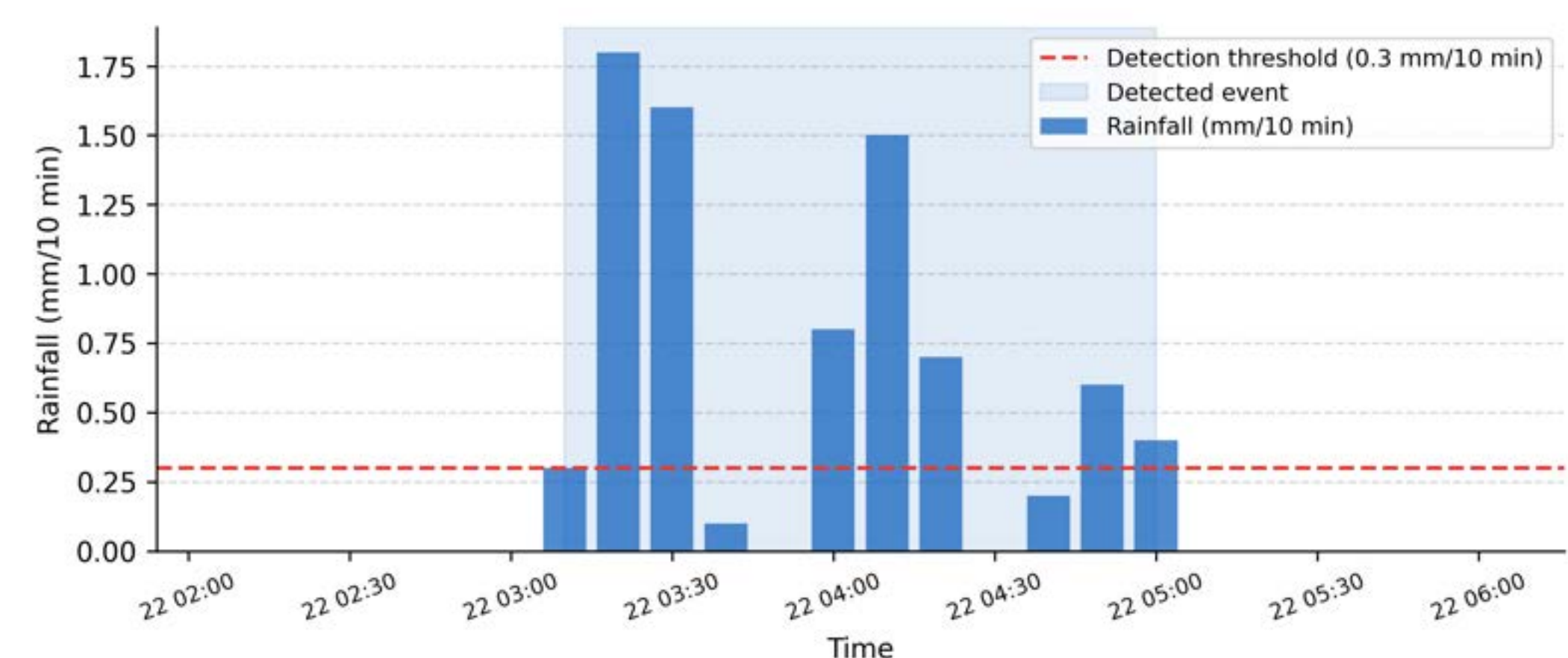


Figure 4: Example of a detected rainfall event at VILLAMORRA station.

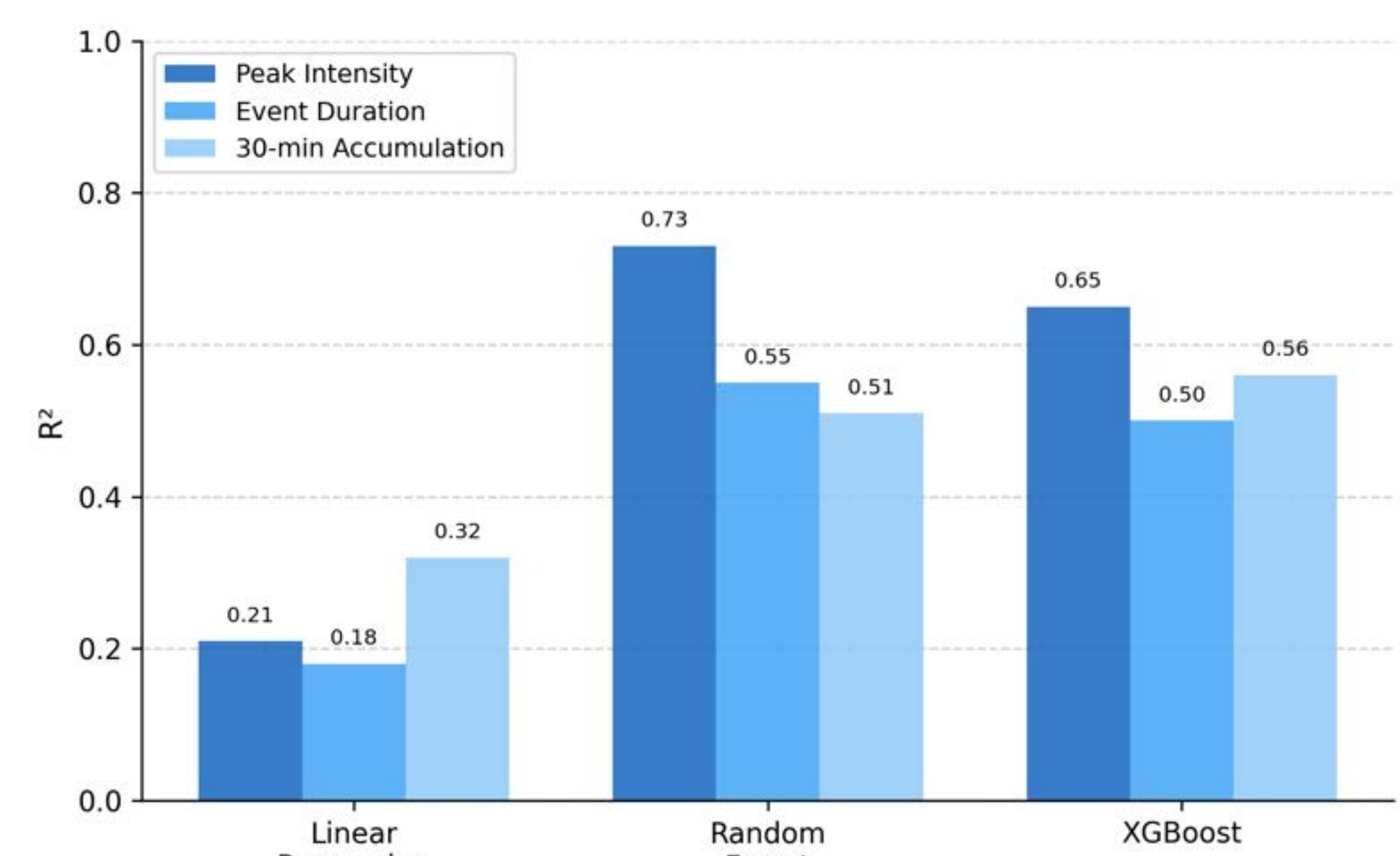


Figure 5: Model comparison across prediction targets.

4. Conclusions

- Tree-based models outperformed linear regression for short-term rainfall prediction.
- Multistation data improved performance by capturing spatial rainfall variability.
- Best results were obtained by XGBoost for accumulation ($R^2 \approx 0.56$) and Random Forest for peak intensity ($R^2 \approx 0.73$).

5. Acknowledgments

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References

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